Rik L.E.M. Ubaghs

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Bio

I am an engineer specializing in neurotechnology and optical systems, with expertise in biomedical imaging, medical devices, and data analytics. My work focuses on translating advanced experimental technologies into practical clinical applications, bridging the gap between cutting-edge research and real-world medical impact.

Education

ETH Zurich, Zurich, Switzerland July 2017 - May 2023
Ph.D. Neurotechnology

ETH Zurich, Zurich, Switzerland Mar 2025 - Mar 2026
CAS Entrepreneurial Leadership in Technology Ventures

University of Amsterdam, Amsterdam, the Netherlands Sept 2014 - June 2016
Research Master Brain and Cognitive Sciences, Cum Laude

Maastricht University, Maastricht, the Netherlands Sept 2011 - June 2014
Bachelor Psychology and Neuroscience, Cum Laude

Skills

Languages:	Dutch (Native), English (Fluent), German (Basic)
Programming:	Python, Rust, Matlab, R
Databases/Querying:	SQL, MySQL
Skills:	Data Science, Biomedical Imaging, Optical Systems, Mechanical Design (Autodesk, Solidworks)

Industry Positions

Arago Labs June 2023 - Current
Co-founder, CEO

- Founded an early-stage medical technology venture focused on improving surgical precision in oncology.
- Designed advanced optical imaging platform combining nonlinear microscopy, and ML/DL-based tissue classification.
- Defined market entry strategy, pricing model, reimbursement pathways, and go-to-market roadmap.

Agora Technology, Maastricht, The Netherlands Nov 2016 - Feb 2018
Founder

- Built data solutions that aimed at streamlining/automating the workflow of medical facilities, especially focusing on surgical equipment.
- We focused on internal logistics, and the creation, maintenance, and analysis of large-scale medical equipment databases using ML/DL.

Research Positions

Inst. of Neuroinformatics, ETH Zurich, Zurich, Switzerland July 2017 – June 2023
Ph.D. Candidate

- Development of a MRI compatible florescence microscope (full-cycle product design).
- Design of a paradigm to combine microscopy and MR imaging methods in awake rodents.
- Design of data pipelines for multi-modal data analysis.

Center for Neuroengineering, Duke University, Durham, USA Dec 2015 – Nov 2016
Visiting Scientist

- Design of a hybrid brain decoding strategy in Non-human Primates.
- Development of ML/DL decoding pipeline for brain decoding.

Spinoza Center for Neuroimaging, Amsterdam, The Netherlands Nov 2014 - Dec 2015
Junior Research Scientist

- MR operator.
- Design and supervision of several large-scale research projects.

Extracurricular Positions

Natural Movements, The Netherlands March 2007 - May 2012
Co-Founder

- Natural Movements is a non-profit organization that aims to engage underprivileged youth in the local community through behavioral coaching and physical education.
- Worked together with social service organisations to provide youth with any additional support they required.

Publications

Publications:

- **Ubaghs, R.L.E.M.**, et al. Simultaneous single-cell calcium imaging of neuronal population activity and brain-wide BOLD fMRI. *bioRxiv* (2023); accepted in *Nature Methods*.
- Rychen, J., ... **Ubaghs, R.L.E.M.**, ... Hahnloser, R. WhaleSonic: Recordings of cetacean vocal behaviour with hydrophone arrays in Northern Norway. *bioRxiv* (2023); submitted to *Scientific Data*.

Patents:

- Sainz Villalba, L., **Ubaghs, R.L.E.M.** DEVICE AND METHOD FOR PROVIDING HISTOLOGICAL NAVIGATION INFORMATION DURING IN-VIVO SURGERY. European Patent Application, Filed 2026.

Honors

Fellowships:

- Pioneer Fellow, ETH Zurich April 2025 - Current

Committees:

- Educational Committee, IIS, University of Amsterdam Sept 2014 - June 2015
- Vice-President Student Council, FPN, Maastricht University Sept 2012 - June 2013
- Program Committee, FPN, Maastricht University Sept 2012 - June 2013